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Meyerian Psychobiology, Personality Development, and the Role of Life Experiences

Michael Rutter, M.D.

The psychobiological life history approach to psychiatry is perhaps the most important of Adolf Meyer's contributions to psychiatric thinking. The author outlines the implications of that approach for current concepts and practice. He reviews the literature regarding the causative role of life experiences in the genesis of psychiatric disorder, the extent to which the effects of stressors are situation-specific, the effects of stress on the organism, the reason for individual variations in the response to stress, and the causes of life stressors.
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Of Adolf Meyer's many contributions to psychiatric thinking, his emphasis on a psychobiological life history approach is probably the most important. Nowadays, I suppose that his psychobiology tends to be thought of in terms of a holistic approach to people as individuals and a rejection of the constraints of traditional disease entities. His alternative proposal that disorders should be viewed as reactions, described in terms of "ergasias," foundered both because of the obscure terminology used and because the ideas seemed too general to be helpful as guides to research or to practice. However, such an interpretation of his concepts misses the central notions that continue to have enduring value. It has been most illuminating in looking back on his writings to realize how far ahead of the time his concepts were and how directly relevant they are now to modern research and current theorizing.

First of all, it should be noted that his conjunction of psychology and biology in the term "psychobiology" had as its cornerstone the need to understand how life experiences had effects on the organism and why there were such large individual differences in people's responses to events and circumstances (1). Particular emphasis was placed on the meaning of happenings for the individual, on their interpretation, on coping mechanisms in terms of both cognitive and affective processes, and on assets and protective factors. He urged the need to take account of "the mental working of the patient and his way of meeting the situation as well as the regulation of his physiological mechanisms" (2, p. 575). He rejected a mental dualism: "Every mental adjustment . . . has its somatic components" (2, p. 585). The requirement was to understand *how* constitutional factors modified responses to events and how the impact of events altered the organism, and to appreciate why life stressors sometimes led only to transient distress and sometimes to a lasting mental disorder. Most of all, Meyer insisted on the *active* role of the individual—the importance of how the experience was thought of and how it was dealt with. He argued that to divest psychology of its dynamic interests was to make it hopeless (2, p. 575); equally he chided psychoanalysts for focusing on the unconscious and on inferred complexes before exploring the facts of how the person thought about and responded to real life experiences (1).

This interactive approach was, of course, not new, nor was it distinctively Meyerian, although he stood foremost among its advocates. He was unusual in his emphasis on the extrafamilial environment, including the school, as a major source of life experiences (3) and in his insistence that development be considered in its social context with attention to the functioning of social systems: "Man functions not only as a 'he' or

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'she' or 'I' but also as a 'they' or 'we', as a . . . member of a group—family, friends or community" (1, p. 86). Most crucially, he insisted that experiences had to be seen not just in terms of their immediate impact but in relation to their effects on personality development and as modifiers of later reactions. Thus, he wrote: "It is always an individual with a history and an outlook in a situation, with the now and a past and a future in action; and we are interested in what we can expect and what *he* expects and what he or she includes" (3, p. 485). He also noted that "an injury leaves . . . scars, but it may also leave an experiential residue, a memory capable of serving in many additional associative possibilities as a person-function, as something to be thought of, remembered, feared and thereby operative in our expectations and motives in life" (1, p. 130).

Throughout his writings on the life history approach he was fascinated by the challenges involved in understanding both the nature of personality and how to conceptualize maturity of functioning, and also the paradoxes presented by the complex intertwining of continuities and discontinuities in the developmental process. He wrote that although personality is "constantly changing, it yet maintains a reasonably orderly internal and external structural and functional organization" (1, p. 6). Also, he observed that "of all the inevitably relative human considerations, maturity is one of the most elusive" (3, p. 426).

Meyer left behind no overarching theory to explain these phenomena, but he did argue for the need to combine imaginative clinical enquiry and rigorous experimental study if we are to gain an understanding of the psychological processes involved in the development of personality and in the genesis of psychiatric disorder. He wrote: "Psychiatry has no dogma to offer. It builds on a willingness to learn from experience. It strives to develop a science of man under the sign of experience and creative experiment" (3, p. 418). What is needed is "a critically trained experimental spirit, chastened and enriched by well trained experience" (1, p. 97). How far has that need been met by the research undertaken in the decades since Meyer expounded the issues to be accounted for? In answering that question I shall be mindful of Meyer's (2) admonition that "no field of biology is more in danger of verbosity than ours . . . let us cultivate conciseness and precision in the statement of what is advanced as new."

THE CAUSATIVE ROLE OF LIFE EXPERIENCES

Any discussion of the role of life experiences in the shaping of personality development and in the genesis of psychiatric disorder must start with the basic question of whether there is any sort of causal influence. Certainly there is ample evidence that there are associations between life events and psychiatric disorder, but do the associations represent causal mechanisms? I do not need to dwell on the difficulties in inferring causation from correlation, but there are ways of using epidemiological/longitudinal data to test causal hypotheses even when controlled experiments are not possible (4).

In drawing conclusions about causative effects, we must draw distinctions between different types of life experiences and different kinds of outcome measures. The findings are reasonably clear-cut that *chronically* adverse experiences can have important deleterious effects on children's behavior and cognitive functioning (5, 6). Within the psychiatric domain, the associations are strongest between family discord or disruption and conduct disorder, especially in boys. The reasons for inferring causation include the results of animal studies showing that experimentally controlled changes in the environment have predictable effects; the demonstration of environment-disorder associations after appropriate control for genetic factors by use of adopted samples, matched groups, or statistical adjustments; the finding that environmental measures predict the later development of psychiatric disorder (7); and the observation that marked environmental change has predictable behavioral consequences. It should be added, however, that the associations are of only moderate strength and that even with the most seriously adverse environments many children do not develop psychiatric disorder.

The evidence linking chronic adversities in childhood with adult psychiatric disorder is open to more methodological objections. Nevertheless, there are good grounds for inferring causal links in the case of the vulnerabilities that follow an institutional upbringing (8) or a marked lack of affectionate care in childhood (9, 10). Serious adversities in the realm of intimate family relationships are associated with increased risks for both personality disorder and depression. Nevertheless, once again individual differences in responses to adversity are great.

There is a large literature on the psychiatric effects of acute life events in adult life. Most reviewers have concluded that events carrying a long-term threat have a truly causal impact in increasing the risk for depressive disorder (11, 12). Doubts have been cast as a result of inconclusive longitudinal studies mostly relying on weak measures (13), but both the analysis of longitudinal data (14) and the detailed study of specific stressors such as unemployment (15) or bereavement (16, 17) suggest that in some circumstances negative experiences *can* provoke the onset of depression. The crucial importance of adequate measures is evident from the demonstrated superiority of systematic interview methods over inventories (18). Two caveats must be added, however. First, once more, individual variation in response is marked; most life events do not lead to depression. Second, there are questions about the importance of acute life events in the genesis of the most severe depressive disorders.

Much less is known regarding the causal influence of acute life events as precipitants of psychiatric disorder in children (19, 20). It seems that such events are associated with psychiatric problems in children, but

the weak temporal association (21) raises doubts about their etiological role when separated from the chronic adversities with which they are so often associated. Also, it may be that the effects of many acute stressors on children's mental health are mediated through their effects on parental mental health and family functioning rather than through any direct impact on the child (22, 23). Nevertheless, it is clear that certain acute changes in life circumstances can have both behavioral and physiological sequelae; this is evident from experimental data in both animals (24) and humans (25). Moreover, the study of events such as the birth of a sibling (26), admission to a hospital (5), bereavement (27), and divorce (28) indicate that life stressors can lead to emotional disturbance. The extent to which they predispose to serious mental illnesses is less certain.

We may conclude that Meyer was right in his supposition that life experiences have an impact on psychological functioning and that in some circumstances they play a part in the genesis of psychiatric disorder.

That conclusion forces us to ask, however, whether the adverse experiences really cause psychiatric disorder rather than understandable reactions to unpleasant events or happenings. After all, it is normal to experience grief following bereavement; why should this be regarded as depression when it happens to be somewhat more severe or persistent than usual? There is no easy answer to that question if only because there are no wholly unambiguous criteria to differentiate "pathological" depression from "normal" distress or dysphoria. Even so, it is clear that chronic adversities are indeed associated with psychiatric disturbances that are much more than simple situational reactions. For example, our own study of children of mentally ill parents (29) showed that the greatest risk applied to disorders that persisted over several years and that were evident in the children's behavior at school as well as at home. It does not seem likely that this was mainly a genetic effect, as marital discord (rather than illness variables) had the strongest association with child disorder within the sample of patients' families. Moreover, the discord associations held strongly even after controlling for the presence of personality disorder in the parents. The same association with persistent disorders applied in our investigation of the psychiatric effects of inner-city stressors. We found that behavioral disturbances were twice as common in children brought up in inner London as in those reared in an area of small towns; the increased psychiatric risk was a function of the higher rate of family adversities in the metropolis. However, the excess of disorder in London children was most marked in the case of disturbances beginning in early or middle childhood and persisting into adolescence (30).

Similarly, Hetherington et al. (28) showed that the children of divorcing parents differed from control subjects in their play and in their behavior at school. Also, Tizard and Hodges (31) showed that children who spent the first few years of life in institutions differed from other children in being more overactive, inattentive, and disruptive at school; moreover, this difference persisted even after adoption into harmonious, well-functioning families. It is clear, then, from much evidence that environmental effects from chronic adversities go far beyond transient situational reactions. Indeed, the effects are sometimes most marked in the case of pervasive and persistent psychiatric disorders.

Does the same apply to the effects of acute life events in childhood? We do not know with any certainty. The studies of major disasters indicate that some children undoubtedly develop socially handicapping emotional disturbances following grossly traumatic experiences (19). In many cases such disturbances are short-lived, but some persist. The same applies to the effects of bereavement (27). In short, the scanty available data suggest that acute stressors most often lead to transient situational disturbances which differ from normal responses only in degree but that occasionally (we do not know how often) persistent psychiatric disorders may result.

A closely related question is whether the disorders that are brought about by acute stressors or chronic adversity persist beyond the period of exposure to the negative experiences. In other words, do the disorders become self-perpetuating and thereby independent from their environmental precipitants? All the available evidence points to the importance of the duration of time over which the adversities operate. This is evident, for example, in the finding that it is the long-term threat associated with acute stressors that mainly accounts for pathogenicity in adults (11). But, in most cases, it is not just the threat that is long-term; the acute stressors lead to a lasting deleterious change in the person's actual life situation. In children, too, it appears that the acute events associated with emotional and behavioral disturbance have their impact by setting in motion persisting changes in environmental circumstances. Thus, the ill effects that are associated with divorce are to a large degree a function of continuing discord or disruption in family relationships (28, 32); those that follow the birth of a sibling, of altered patterns in parent-child interaction (26). To that extent, the distinction between acute stressors and chronic adversities is somewhat artificial; both concern persistent experiences. Nevertheless, there is a difference in that some of the acute life events seem influential because they lead to a loss of a love relationship or the loss of self-esteem; that is not necessarily the same experience as not having a love relationship in the first place.

However, is it correct to see all pathogenic stressors as involving a lasting change in the environment? We do not know for sure. Are the ill effects associated with recurrent hospital admission in childhood (33), for example, explicable in this way? It may be relevant, perhaps, that there are no detectable ill effects from just one admission—the adverse sequelae follow re-

peated admissions. Does this mean that the consequent psychological disturbances are largely a consequence of, say, altered patterns of parent-infant interaction, as is the case with experimentally induced separations in monkeys (34)? We lack the data to answer that question—a lack that highlights the need for greater study of the ways in which acute stressors operate.

Nevertheless, even if we assume that in many instances acute life events exert their damage as a consequence of a lasting environmental change, the question remains: Do the disorders persist if environmental circumstances change for the better? Of course, much will depend on how much circumstances improve. Certainly, there is every reason to suppose that individuals with psychiatric disorders remain susceptible to environmental influences. Marked environmental change for the better is accompanied by psychological benefits (6, 35). Even so, it is striking that in many cases child psychiatric disorders do persist in spite of environmental improvements. This was evident, for example, with respect to reductions in family discord both in our own study of families in which the parents had a mental disorder (29) and in Richman et al.'s (7) general population longitudinal study. Of course, that is not to say that when the discord diminished there were not other, perhaps more subtle, family adversities that remained. But it is clear that it would be a misleading oversimplification to regard the psychiatric disorders as just immediate responses to stress and adversity. That conclusion alone serves as a warning that it is not likely to be useful to classify mental conditions simply into those due and those not due to stress.

WHAT HAPPENS TO THE ORGANISM AS A RESULT OF STRESS OR ADVERSITY?

The question then arises of what happens to the organism as a result of stress or adversity (36). If the disorder is not merely a situational reaction, what causes the effects to persist? It is important to appreciate at the outset that somatic changes as a result of stressful experiences are well established. In animals, for example, it is well-known that acute physical stresses can lead to lasting structural and functional changes in the neuroendocrine system (37). In humans, psychological stressors, too, have been shown to lead to altered patterns of hormonal activity such that the response to stress situations which have been encountered repeatedly in the past is different from that evident on the first occasion (38). Gross restriction in sensory input leads to changes in the brain (39–41). In humans, the practical effects are most obvious in the case of the loss of binocular vision that stems from an uncorrected strabismus in infancy (42, 43). Less certainly, there may be somewhat comparable effects on verbal learning associated with severe but transient hearing loss during the early years (44). We may accept that adverse experiences of various kinds can, under certain circumstances, lead to lasting somatic changes. However, it seems rather unlikely that such changes account for most of the psychiatric disturbances that follow stress or adversity. Are there other types of change that could be responsible for persisting patterns of emotional or behavioral disturbance? Several rather different possibilities are apparent.

To begin with, there may be altered patterns of responsiveness to the environment. Thus, Dodge (45) found that aggressive boys were more likely than other boys to attribute hostile intentions to others; as a result, they elicited as well as initiated more negative interactions. Similarly, Patterson (46) found that parents who experienced coercive family interactions with aggressive sons were more likely to perceive neutral behaviors as deviant and more likely to develop a general attitude of rejection toward their sons. The family as a whole acted in ways that prolonged negative interchanges, and the aggressive boys were less susceptible than normal to discipline (47). Of course, in these studies it is not possible to separate the effects of experiences from constitutional predispositions, but the circumstantial evidence suggests that adverse experiences can lead to changes in the way that people respond to their environment. The findings also suggest that in part this is due to people's feelings, thoughts, or interpretations regarding what has happened to them. As Kagan (48) argued, "The person's interpretation of experience is simultaneously the most significant product of an encounter and the spur to the next" (p. 279).

One specific aspect of responsiveness that requires emphasis is an individual's sense of self-esteem and self-efficacy—the recognition that to an important extent one is in control of one's destiny and can deal with the life challenges that are encountered. It has often been noted that it is this conviction which is most strikingly lacking in people who have suffered serious prolonged adversities; all too often their repeated suffering has led them to feel at the mercy of fate and hence to respond passively and ineffectively to new stressors or indeed to new opportunities. For example, in our follow-up study of institution-reared women (8) we found that they were twice as likely as control subjects to marry for negative reasons, to escape from an intolerable situation but without planning or foresight as to whether their action would improve or worsen their situation. Not surprisingly, many of these unplanned marriages were to deviant men from similarly disadvantaged backgrounds, and a high proportion of the marriages broke down. In this case, the way in which girls dealt with one stressor (an unhappy family situation) created another (a discordant, disharmonious, unsupportive marriage). The continuities in psychological problems may have stemmed initially from a damaged self-concept, but later they derived from partially self-created environmental adversities. Some of the reasons for persistence derive from the ways in which alterations in the individual create further environmental hazards.

One aspect of this process of person-environment interaction concerns the effects of life events on patterns of interpersonal relationships. For example, Hinde and McGinnis (34) showed experimentally that separation experiences in rhesus monkeys intensified insecurity in parent-infant attachments and that it was the poor parent-infant relationship that was associated with the persistence of emotional disturbance in the infant. Somewhat similarly, Dunn and Kendrick (26) found that the birth of a sibling was associated with changes in the mother's style of interaction with the firstborn. It should be emphasized that it is not just a matter of one type of relationship generalizing to another. Indeed, girls who had an intense playful relationship with their mothers were more likely to have a poor relationship with their younger siblings. Perhaps the need to share an affectionate relationship increased the rivalry and jealousy. Perhaps, too, it is these broader changes in patterns of family interaction that explain in part why sibling jealousies are so surprisingly persistent. At first sight it is odd that they should last, in that the developmental changes are so marked; the neonate provides a very different "stimulus" for relationships from, say, an active, verbal 4-year-old. Yet the sibling relationship in the newborn period shows a considerable degree of continuity with that evident 4 years later (49)—a continuity substantially greater than that found for individual temperamental attributes (50), for example. It is possible that the continuity lies more in the persistence of altered patterns of family interaction than in any changes in the individual as such.

A further mechanism is to be seen in the effects of a stress experience in altering a person's response to some later stressor, an effect that may enhance or reduce adverse reactions—that is, stress may "sensitize" or "steel" (25). It is not at all clear just what determines which reaction occurs or, indeed, what psychological or physiological processes are represented by the concepts of "sensitization" or "steeling." It may be that, in part, it is a matter of attitude or frame of mind. For example, perhaps that is the reason why young children who are used to happy separations (51) or who are well prepared for hospital admission (52, 53) are less likely to show emotional disturbance. After all, it is clear that how we view an experience may determine whether it is positive or negative in its effects—whether a life change is a challenge and an opportunity or a burden and a strain. A degree of fear in the right circumstances is exciting and pleasant (as shown by one's enjoyment of roller coasters or mountain climbing), but in other circumstances it is wholly unpleasant and severely stressful. It also may be that coping mechanisms are important. Many life events require action to deal with them effectively, and people vary in how well they cope. Success in coping may be influenced by past experiences. For example, we found that institution-reared women functioned as well as did control subjects in early adult life provided they had an emotionally supportive husband and reasonable living conditions, but when faced with adverse circumstances in adult life they were twice as likely as the control subjects to succumb (8). It seemed that in some fashion their experience of serious adversities in childhood had left them less prepared to deal with later life hazards.

Thus far, I have focused on the effects of stress and adversity in terms of a variety of mechanisms that create indirect, rather than direct, long-term effects. That is indeed what the evidence suggests; the notion that adverse experiences lead to lasting damage to personality "structure" has very little empirical support (5, 6, 36). However, it is possible that in some circumstances serious and prolonged early adversities may result in impaired socioemotional capacities. The main contender for such an effect is provided by the lack of opportunity to develop selective bonds or attachments during the preschool years, as may be the case with some types of institutional care with multiple, changing caretakers. For example, Tizard and Hodges (31) found that 8-year-old children who, as infants, had been raised in residential nurseries and then adopted late still tended to show inattentive, poorly modulated social behavior at school. Most of these children had developed close ties with their adoptive parents, so the problem was not an inability to form selective love relationships; nevertheless, their peer interactions and their response in group teaching situations remained maladaptive. A further follow-up to age 16 (Hodges and Tizard, 1985 unpublished data) showed some attentuation of the effects of early experiences but also a surprising degree of persistence in the difficulties with close peer relationships more than a decade after adoption into harmonious, well-functioning families. All the evidence suggests that changes in personality functioning occur right into adult life, but it may be that certain types of experience impair personality development in a manner that is intrinsic to the individual. If so, it is likely that the effects are relative rather than absolute. Longitudinal data all indicate the modifiability of human functioning provided that the environmental conditions are sufficiently good.

WHY IS THERE SO MUCH INDIVIDUAL VARIATION IN RESPONSE?

The considerations that I have just discussed concern the question of why the ill effects of stress and adversity sometimes endure. However, the fact is that not only are the sequelae often transient rather than persistent, but also there are huge individual variations in the response to environmental hazards. Some people succumb; some escape unscathed. Accordingly, I need to turn now to the issue of why there is so much individual variation in response.

The first basic point to make is that many of our measures of life events provide only indirect indicators of the actual experiences that create the psychological

hazard; to a large extent, the apparent individual variation in response is a consequence of noncomparability of experiences. For example, Brown et al. (9) have shown that the long-term vulnerabilities associated with early parental loss derive not from the loss per se but from the lack of affectionate care to which the loss may lead. Similarly, we found that the psychiatric risks for the children of mentally ill parents mainly stemmed from the accompanying family discord rather than from the parental illness as such (29). Also, Hetherington et al. (28) observed that the adverse psychological sequelae of parental divorce were largely a function of the disturbances in family functioning and relationships that occurred. The findings serve to remind us of the importance of the distinction between risk indicators and risk mechanisms; they also alert us once more to the evidence that to an important extent the long-term effects of acute experiences are a result of the chronic adversities to which acute events give rise.

There is also evidence that the meaning of the event for the individual may be relevant. For example, it has been suggested (54) that events which are dissonant with normal life cycle expectation may be more stressful than those which are "normative" in their timing. The empirical data to test this hypothesis are extremely limited, but there is some modest support for the idea. Thus, it has been suggested that psychiatric sequelae are more likely when bereavement stems from the death of a child or from suicide rather than from death in old age (17); similarly, unemployment appears to be more stressful when it occurs in mid-life than in either the teen-age period or the years just before normal retirement (15).

The personal meaning of events may also be relevant. Hetherington et al. (28) found that a mother's having a job outside the home was not ordinarily a risk factor for her children. However, when the mother first went out to work at the time of a divorce, the children were more likely to show disturbance; it seemed that the coincidence of paternal absence through divorce and maternal absence through work made maternal employment have a "loss" element that was not present in other circumstances. Similarly, Brown and Bifulco (55) have argued that life events are more stressful when they reflect some area of role conflict. Also, there is some suggestion that the psychiatric risks associated with parental death or mental illness may be greater for a child of the same sex (29).

The social context in which life experiences occur is another important consideration. First, as I have already noted, life events may exert their main effect through their bringing about perturbations in social interactions and relations. This is evident, for example, in the consequences of the birth of a younger sibling (26). Second, the level of psychological demands on an individual is greatly influenced by the social context. Thus, Schaffer and Liddell (56) showed that adults were more likely to ignore children's bids for attention in polyadic than in dyadic situations; their speech also was somewhat more likely to be peremptory and prohibitive. Sensitive responsiveness in parenting is a function of the situation as well as of individual personalities and dyadic meshing. Perhaps for this reason we found (8, 57) that the deficits in the parenting of institution-reared women were more evident in real-life parenting breakdown in the context of multiple social stresses than in the mother-child interaction when the two were at home together at a time when there were no competing demands.

Third, the social context may determine the appraisal of events. This is most strikingly shown in the wartime studies demonstrating that acute combat involving great danger may produce no evidence of stress, either behavioral or hormonal (58, 59). It appears that in some circumstances job prestige and high group morale may be sufficient to counteract the effects of severe physical danger.

Fourth, there is substantial evidence that psychiatric disorder is much less likely to occur when adults have the support of a harmonious, loving marital relationship (8, 11, 60, 61). Other social supports, too, are associated with protection against mental disturbance (62). The mechanisms involved in these associations have proved quite difficult to delineate, however, and it is clear that the key factor is not just the availability of social contacts or the frequency of social interactions; rather, it is the perceived adequacy and psychological use made of the social support (63, 64). This finding raises the question of whether inadequate support is a function of the individual personality, the environment, or the person-environment interaction. It also reflects the observation that a high level of social support is not necessarily benign; social dependence can sometimes create tension (65). A further issue is whether the availability or lack of social support has a direct effect on psychological functioning or whether it serves as a buffer or modifier that alters the impact of other stressors. There are both conceptual and statistical problems in testing for such a buffering effect, but it seems likely that in some circumstances it occurs (66, 67), although it is doubtful whether social supports act only in this way.

The buffering hypothesis also serves to draw attention to the need to pay attention to possible protective factors that serve to ameliorate the effects of stress (67). The study of resilience in the face of adversity is of very recent origin and there is a paucity of relevant data. Nevertheless, such evidence as there is suggests that the protective factors promoting resilience include two key features: secure, stable affectional relationships and the experiences of success and achievement. It appears that with both sets of factors the positive elements need not necessarily be global. One good close relationship does much to reduce the damage from discord and disharmony in other social interactions; lasting rewards and achievements in one domain of activity may offset failures in other arenas of life. However, it probably is necessary for the successes to occur in areas that the individual regards as important.

An unresolved issue in the field of stress research is the question of whether life events summate and balance (68). The Holmes and Rahe (69) approach assumes that life events are additive in their effects (that is, the impact of several undesirable events is greater than just one); in contrast, Brown and Harris (11) have argued that just one event with a long-term threat is sufficient to precipitate psychiatric disorder and that it matters little if there are other events. We lack adequate evidence to decide between these two views. In addition, there is the further question of whether positive life experiences serve to balance negative ones—for instance, whether the impact of a personal loss will be less if it is accompanied by a promotion at work or the making of a new friendship. The notion that the balance between pleasant and unpleasant events may be psychologically important is plausible (70–72), but, again, adequate data to test the hypothesis are lacking. Intuitively, it seems likely that it may be necessary for the content of the positive event to "match" that of the stressful happening if protection is to occur. That is, the making of a new love relationship may mitigate the loss of an old one, whereas the winning of a race or a prize may not. The possibility has yet to be investigated.

Last, of course, the explanation of individual variation in responses to stress and adversity must be sought in the characteristics of the individuals themselves. A person's sex is relevant in that connection: boys appear to be more liable than girls to suffer ill effects from family discord and disruption (73), and marital breakdown seems to have a more detrimental effect on men than women (74), although sex differences in responses to marital dissolution appear to vary according to the health measure used (75). The reasons why boys are more psychologically vulnerable than girls in some respects are poorly understood. In part, it is likely to be because young males elicit more damaging behavior from others; thus, Hetherington et al. (28) found that parents more often quarreled in front of their sons than their daughters, and Ruppenthal et al. (76) found that socially deprived monkeys were more likely to abuse and kill their male than their female infants. In part, too, it may be because boys respond to adversity in ways that are less likely to enlist support and care from others. However, it also is possible that males have a greater inherent psychological susceptibility in a fashion comparable to their greater physiological vulnerability to physical hazards (77).

Temperamental variations, as well as sex, serve to modify responses to stress and adversity (78). Dunn and Kendrick (79) found that features such as malleability, emotional intensity, approach/withdrawal with respect to new situations, and preponderant mood were associated with children's responses to the birth of a younger sibling in terms of both emotional upset and pattern of mother-child interaction. Other investigations have found that temperamental attributes were associated with variations in children's responses to parental mental disorder (80, 81), institutional care (82), delinquency risk factors (83), and, possibly, even malnutrition (84). It should be noted, however, that the particular temperamental features that are protective vary according to circumstances. The mechanisms involved in this mediating role of temperament are both direct and indirect. Thus, Kagan et al. (85) showed the importance of individual differences in physiological reactivity in relation to preschool children's behavioral inhibition—an example, probably, of a relatively direct effect. We found (80, 86) that within families with a mentally ill parent, children with "difficult" temperaments were more likely to be the target of parental criticism, irritation, and hostility—an indirect effect.

Coping mechanisms are likely to be important, too, in determining whether or not people develop psychological disturbance following stress or adversity (25). This seems likely if only because many stressful life experiences pose challenges or require adaptations, and it would seem probable that the outcome should be modified by how people deal with such crises. However, whether or not it is fruitful to seek to classify coping mechanisms according to their inherently adaptive or maladaptive qualities is much more dubious. What is best in the circumstances may vary according to the nature of the life event and according to the individual. Nevertheless, it does seem to matter that people make some appropriate response—to act and not simply react—as shown, for example, by Brown et al.'s (9) findings on how girls deal with an unwanted pregnancy and by our own data (8) on how young women plan their work and their marriages.

There are many different reasons why people vary in their vulnerability to adverse experiences. Genetic factors may operate in part through their determination of individual differences in susceptibility to particular environmental factors (87, 88). Past experiences may also be important through their role in creating vulnerability or resilience to current stressors. As already noted, certain seriously adverse experiences in childhood such as an institutional rearing (8) or a prolonged lack of affectionate care (9) have been shown to predispose one to adult psychiatric disorder. The precise processes whereby this vulnerability arises and the particular mechanisms by which the vulnerability leads to psychiatric disorder are not known. However, it has been suggested that one way is through the creation of a self-concept or cognitive style that makes successful coping with life stressors less likely (11).

WHAT CAUSES LIFE STRESSES?

The last issue to be considered is why people differ in their life experiences and how variations in environmental circumstances arise. In other words, what causes life stressors and adversities? Of all the topics covered in this paper, this one has been the least investigated up to now but, in many respects, it is the

most crucial. One of the important methodological advances in life events research was the recognition that it is necessary to separate events that might have been brought about by the mental disorder from those that were independent (68, 89). For obvious reasons, the causal hypothesis could be tested adequately only through the use of independent events. However, that leaves several crucial matters unaddressed. To begin with, the notion of independence from the illness is by no means the same as independence from the person. Some investigators (11) have largely restricted the concept to the illness, but this allows all sorts of events to be classified as "independent" when in fact the person's own behavior may have played a role in bringing the event about. For example, a spouse's infidelity or a son's arrest, although obviously not due to the patient's illness, may well have been influenced by family discord and difficulties in which the patient played a key role. Loss of a job when a firm closes down might seem to be independent from both the illness and the person in that the firm closure is clearly outside the individual's control. But, still, we must ask why he or she was in that job rather than another. Choice of type of job is influenced by personal characteristics deriving from both experiential and constitutional factors. Of course, that is not to argue that societal factors are not also of major importance in employment. Obviously, both job opportunities and selection are hugely influenced by socioeconomic and political forces (90). The point being made is that against that background of powerful external pressures there is still individual variation in how people respond—a variation that has an influence on outcome. Thus, delinquent and behaviorally disturbed boys are more likely to leave school early without taking exams and so land in less skilled occupations (91, 92). Similarly, both longitudinal (18) and retrospective (9) studies of women have shown that serious adversities in childhood are associated with an increased likelihood of a woman's marrying a man with an unskilled or semi-skilled job. This is relevant to people's experiences of acute stressors and chronic adversities in adult life, as both are more frequent in working-class groups (11).

It is clear that if a broad concept of independence is used, many life events can be regarded as only "possibly independent." Costello (93), unlike Brown and Harris (11), found that possibly independent events were more likely to precipitate depression than were definitely independent ones and argued that both the stressors and the depression might derive from personality variables. Indeed they might, but it should be added that the personality style might stem from earlier adverse experiences as well as from constitutional influences. It is clear that this possibility raises crucial and difficult methodological problems in the testing of causal hypotheses. Nevertheless, it is important to appreciate that life events brought about by a person's own actions may still serve as a truly causal influence in the genesis of psychiatric disorder. This is most easily illustrated by reference to physical causes of medical conditions. Thus, there is no doubt that cigarette smoking is a causal agent for lung cancer (94), but smoking is a voluntary action determined by personality variables and environmental influences (95, 96). The fact that smoking is brought about by the person's own actions does not mean that smoking is not involved in the genesis of lung cancer. Similarly, impulsive, overactive, behaviorally disruptive children are more likely to experience head injuries just because their behavior leads them into risky situations so that they suffer accidents from falling off roofs or out of trees (97). Nevertheless, however brought about, severe brain damage truly increases the risk of psychiatric disorder, as shown by the changes over time following head injuries (97). Exactly the same could apply to psychosocial stressors and chronic adversities. The need for rigorous testing of causal hypotheses remains, of course, but it is important that we get away from the absurd dualism that supposes that causes are either internal or external but cannot be both at the same time. They can, and frequently are. This is well illustrated, for example, by the empirical demonstration that just as sex hormones influence behavior, so also behavior modifies sex hormone levels (98); also by the evidence that the biochemical changes associated with mania can be brought about by a patient's pretending to be manic (99).

The issue of what causes life stressors is not just a methodological quibble over the criteria for independence; the question is central to the study of life events. If acute stressors and chronic adversities play a role in the causation of psychiatric disorder, we must ask why it is that some people have more adverse experiences than do others. Is it chance or ill luck? Or, rather, are the negative experiences explicable in terms of individual characteristics or past adversities? Insofar as they are due to individual features, are these ones that are linked genetically with hereditary influences that predispose to depression? It is now appreciated that to an important extent, genetically determined qualities can serve to create environments (100). Equally, it is evident, however, that adult functioning is shaped in part by chains of indirect environmental linkages deriving from adversities in childhood (36). Research has only just begun to tackle these questions, but they constitute central items on the agenda of investigations that need to be undertaken.

CONCLUSIONS

In concluding, I need to return to Adolf Meyer. In my introductory remarks I noted the ways in which he laid out the concepts and issues involved in a psychobiological approach to the role of life experiences in personality development and in the genesis of psychiatric disorder. Subsequent research and contemporary theorizing (87) have shown how right he was in the aspects that he chose to emphasize.

Acute stressors and chronic adversities do indeed make an important impact on psychological functioning. However, they impinge on an active, not a passive, organism. The effects of life experiences are much influenced by how individuals appraise their situation and by how they respond to the challenges and adaptations involved. The very marked individual differences in reaction have their origin in constitutional variations in susceptibilty, in vulnerabilities and resilience created by earlier experiences, in the presence of protective influences, in the social context of the negative experiences, and in the particular characteristics of the person-environment interaction. It is evident that experiences can change the organism; it is equally clear that people create their environments. This two-way interaction constitutes the pivot of the central concepts of psychobiology. Adolf Meyer clearly outlined the questions that needed to be tackled; now that we too are convinced of the importance of the issues, we need to turn our minds to the question of *how* to disentangle and delineate the processes and mechanisms involved. Obviously, the answer is not to be found in any one research design but rather in an integration of different investigative strategies. Some useful progress has been made in that direction, but it is evident that there is still a long way to go. What research has shown is that the journey should prove well worthwhile.

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